Historical Instruments and Exclusion Restrictions: A Replication Study of Roche (2020)*

Daehee Kim
Sogang University
daeheekim@sogang.ac.kr

Sungwon Lee†
Sogang University
sungwonlee@sogang.ac.kr

Jiho Yeom
Sogang University
ygh0902@sogang.ac.kr

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Abstract

This paper conducts narrow and wide replications of Roche (2020), which uses historical instruments to estimate causal effects, and evaluates the validity of the exclusion restrictions for the instruments. We rely on the framework of Masten and Poirier (2021) and estimate the falsification frontier and falsification adaptive set. We find that the baseline model, which imposes exact exclusion on both instruments, is falsified, and that the original sign conclusion of Roche (2020) is not preserved over all minimally non-falsified relaxations of the exclusion restrictions. However, it is maintained over a large portion of the relaxations. The recovery of the sign conclusion depends on which instrument is maintained as valid. In our preferred specification, the estimated effect is positive and similar in magnitude to that in Roche (2020). In doing so, we illustrate how sensitivity analysis of the exclusion restriction can be conducted, including how the robustness of a conclusion depends on which instrument is maintained as valid.

Keywords: instrumental variables; exclusion restriction; historical instruments; replication; partial identification; falsification adaptive set; breakdown frontier.

JEL Codes: C26, C36, C52, O31.

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†Corresponding author: 35 Baekbeom-ro Mapo-gu, GN710, Seoul 04107, South Korea.

1 Introduction

Historical variables are widely used as instrumental variables (IVs) in empirical economics (e.g., Acemoglu et al. (2001); Baum-Snow (2007); Tabellini (2010); Duranton and Turner (2011); Duranton et al. (2014); Chyn et al. (2026)). They are often used to isolate variation in institutions, geography, infrastructure, culture, technology, or local economic conditions. Such variables are useful as instruments because they can provide exogenous variation in the endogenous variable arising from their predeterminedness.

However, historical variables may violate the exclusion restriction, because historical forces can affect current outcomes through multiple persistent channels. The literature commonly argues that historical instruments may capture broad historical development paths rather than a single causal mechanism (e.g., Bazzi and Clemens (2013); Casey and Klemp (2021)).

Roche (2020) provides a setting in which exactly this concern arises. Roche (2020) estimates the effect of road connectivity specified as street density on innovation in terms of the number of patents. However, road connectivity may be endogenous, as there are possible omitted variables correlated with both road connectivity and the number of patents, such as urban growth and location fixed effects that attract people to gather and enhance innovation. To resolve the potential endogeneity issue, Roche (2020) considers two IVs: the percentage of housing units in a given block group that were built before 1940 and the percentage built between 1940 and 1949, denoted by HUpre1940 and HU1940s, respectively.

There may be several channels through which these variables have direct effects on the outcome variable, thereby violating the exclusion restriction. One such channel is a combination of the persistence of the spatial distribution and agglomeration. It is well documented in the literature that historically dense regions tend to remain dense and urban development can be persistent over time (e.g., Davis and Weinstein (2002); Glaeser and Gyourko (2005)). In this context, a higher share of early housing—whether built before 1940 or during the 1940s—is likely to characterize neighborhoods developed relatively early in the urbanization process. As a result, a relatively high share of housing units built before 1940 and during the 1940s is likely to be strongly correlated with the contemporary share of housing units in each block group. In addition, urban density is positively associated with knowledge spread via agglomeration (Carlino et al. (2007); Arzaghi and Henderson (2008); Lin (2011); Kantor and Whalley (2014), to name just a few). Taken together, the persistence of spatial density and its agglomeration ben-

efits imply that a high historical housing share may raise current innovation directly through agglomeration, independently of road connectivity—a direct channel that would violate the exclusion restriction.

In this paper, we replicate the results in Roche (2020) in both the narrow and wide senses. For the wide replication, we apply the partial identification framework of Masten and Poirier (2021), relaxing the exclusion restrictions continuously through sensitivity parameters. We test whether the baseline linear IV model is falsified; when it is, we estimate the falsification frontier (the minimal relaxations under which the model is just non-falsified) and the falsification adaptive set (the parameter values consistent with those relaxations).

Based on the estimated falsification frontier, we consider which minimally relaxed assumptions recover the main conclusion of Roche (2020). To this end, we introduce the notion of a "*conclusion-preserving region*," the subset of the falsification frontier on which the sign conclusion of Roche (2020) is preserved, combining the falsification frontier of Masten and Poirier (2021) with the breakdown frontier of Masten and Poirier (2020).

We successfully replicate the results in Roche (2020) in the narrow sense. Turning to the wide replication, we show that the baseline IV model is falsified and estimate the falsification frontier. Our estimated falsification adaptive set contains both positive and negative values, although the point estimate of Roche (2020) is included in the set. This indicates that the original sign conclusion of Roche (2020) is not preserved across *all* minimal relaxations of the exclusion restrictions consistent with non-falsification. The estimated conclusion-preserving region, however, shows that the main conclusion of Roche (2020) is preserved over a large portion of the falsification frontier, indicating that her main result is robust to these relaxations. Moreover, in our preferred specification—using HUpre1940, the instrument we argue to be more credibly excluded, as the excluded instrument and including HU1940s as a control—the estimated effect remains positive and statistically significant, and is similar in magnitude to that in Roche (2020).

This paper illustrates how sensitivity analysis of the exclusion restriction can be conducted in applied work when multiple instruments are available for a single endogenous treatment. In particular, we show how our sensitivity analysis can be used to evaluate the robustness of a conclusion, including how that robustness may depend on which instrument is maintained as the valid exclusion.

The rest of this paper is organized as follows. Section 2 describes the data and introduces

the framework of Masten and Poirier (2021), and Section 3 reports the wide replication results. Section 4 concludes. The narrow replication results, together with some additional results, are provided in the Appendix.

2 Data and Empirical Framework

We use the original data from Roche (2020). The unit of analysis is the block group. As a measure of physical connectivity, Roche (2020) defines connectivity as the total miles of multimodal streets in a block group in 2010, divided by the block group's area in square miles. Using the original data and specification, we estimate the following block-group-level equation:

$$Y_i = \alpha + \theta D_i + X_i'\beta + U_i, \tag{1}$$

where Y_i is the number of granted patents applied for between 2011 and 2013 by assignees located in block group i, D_i is road connectivity, X_i is a vector of control variables, and U_i is an unobserved error term. The parameter of interest in (1) is θ.

Connectivity is unlikely to be exogenous. As Roche (2020) argues, a block group's physical connectivity is plausibly correlated with unobserved determinants of innovation—for instance, local amenities that draw skilled workers, or favorable initial conditions for development—so that ordinary least squares would not recover the causal effect θ. To address this, Roche (2020) instruments connectivity with two historical housing variables, HUpre1940 and HU1940s. The corresponding first-stage equation is

$$D_i = \pi_0 + \pi_1 Z_{1i} + \pi_2 Z_{2i} + X_i'\gamma + V_i, \tag{2}$$

where Z_1 and Z_2 denote the two historical instruments (HUpre1940 and HU1940s), X_i is the same control vector as in (1), and V_i is an error term independent of Z_i and X_i. The baseline model maintains that both instruments satisfy the exclusion restriction.

Because the instruments are themselves historical variables, however, their validity is not beyond question: the same unobserved determinants of innovation—in particular, favorable initial conditions for development—may also be correlated with the historical composition of the housing stock. Whether, and to what extent, this compromises the identifying assumptions is precisely the question our analysis takes up.

We relax the exclusion restrictions using the framework of Masten and Poirier (2021). To this end, consider the classical linear model with multiple instruments. Let $Y(d,z)$ denote the potential outcomes defined for each $(d,z) \in Supp(D,Z)$. Following Masten and Poirier (2021), we specify the following linear model for the potential outcome:

$$Y(d,z) = d\theta + z'\gamma + U,$$

where $\left(\theta, \gamma'\right)' \in \mathbb{R}^{1+l}$ is a constant vector with $l = \dim(z)$, and U is an unobserved error term. In this paper, $l = 2$ as there are two IVs. Roche (2020) incorporates a rich set of covariates, denoted by X. Accordingly, in this paper, the main variables Y, D, and Z can be viewed as residualized versions of the original variables after controlling for X. We impose the following assumptions:

Assumption 2.1. $rank\left(Cov\left(Z,D\right)\right) = 1.$

Assumption 2.2. $Var(Z)$ *is invertible.*

Assumption 2.3. $Cov\left(Z,U\right) = 0.$

Assumption 2.4. *There are known constants* $\delta_l \geq 0$ *such that* $|\gamma_l| \leq \delta_l$ *for* $l = 1,2.$

Assumption 2.1 is the standard relevance condition for Z. Assumption 2.2 requires that the IVs have sufficient variation. Assumption 2.3 is the standard exogeneity condition for Z. Assumption 2.4 is referred to as the partial exclusion restriction, which is also considered by Conley et al. (2012). It requires that the magnitude of the direct effect of instrument Z_l be bounded by δ_l. Since the exclusion restriction requires $\gamma_l = 0$ for all $l = 1,2$, setting $\delta_l = 0$ for all $l = 1,2$ is equivalent to imposing it. Consequently, increasing δ_l corresponds to a relaxation of the baseline IV model with respect to Z_l.

Note that the partial exclusion restriction does not imply that the exclusion restriction is necessarily violated. Rather, it permits a violation of the exclusion restriction while bounding its magnitude. The larger δ_1 is, the larger the direct effect of Z_1 on the outcome variable is permitted to be. Furthermore, $\delta \equiv (\delta_1, \delta_2)'$ allows us to consider a continuous relaxation of the exclusion restriction separately for each instrument. For example, setting $\delta_1 = 0$ while varying δ_2 can be understood as imposing the exclusion restriction on Z_1 and allowing a violation on Z_2. This asymmetric structure, in which some instruments remain valid while others are only

partially excluded, is central to the sensitivity analysis, where we trace the values of δ at which the model becomes falsified.

Masten and Poirier (2021) show that under Assumptions 2.1–2.4, the identified set for θ is given by

$$\Theta(\delta) = \left\{ b \in \mathbb{R} : -\delta \leq Var(Z)^{-1}(Cov(Z,Y) - Cov(Z,D)b) \leq \delta \right\}, \tag{3}$$

where the inequalities are interpreted componentwise.

Let $\psi \equiv Var(Z)^{-1}Cov(Z,Y)$ and $\pi \equiv Var(Z)^{-1}Cov(Z,D)$. Proposition 2 in Masten and Poirier (2021) characterizes the falsification frontier FF, when $\dim(D) = 1$, which is directly applicable to our case:

$$FF = \left\{ \delta : \delta_\ell = |\psi_\ell - b\pi_\ell|, \quad \ell = 1,2, \quad b \in \left[\min_{\ell=1,2 \,:\, \pi_\ell \neq 0} \frac{\psi_\ell}{\pi_\ell}, \max_{\ell=1,2 \,:\, \pi_\ell \neq 0} \frac{\psi_\ell}{\pi_\ell} \right] \right\}, \tag{4}$$

where ψ_l and π_l are the l-th components of ψ and π, respectively, for $l = 1,2$. The FF is the set of values of δ that are not falsified but that would be falsified if any component were strengthened.[1]

The falsification adaptive set FAS is defined as the union of the identified sets over δ on FF; i.e., $FAS = \cup_{\delta \in FF} \Theta(\delta)$. Theorem 2 in Masten and Poirier (2021) provides the closed-form expression of the FAS as follows:

$$FAS = \left[\min_{\ell=1,2 \,:\, \pi_\ell \neq 0} \frac{\psi_\ell}{\pi_\ell}, \max_{\ell=1,2 \,:\, \pi_\ell \neq 0} \frac{\psi_\ell}{\pi_\ell} \right]. \tag{5}$$

The FAS summarizes the parameter values consistent with the minimally non-falsified relaxations. If the FAS contains negative values, then the sign conclusion $\theta \geq 0$ is not guaranteed to hold across all minimally non-falsified relaxations. Estimating the lower and upper bounds of the FAS requires estimating ψ_l/π_l for each $l = 1,2$. These ratios can be estimated by 2SLS, using one instrument (i.e., l-th instrument) as the excluded instrument and the other as a control variable (Lemma 1 in Masten and Poirier (2021)).[2] Note that the estimators are obtained by partialling out the exogenous covariates X from Y, D, and Z (i.e., the Frisch–Waugh–Lovell

[1] Since smaller values of δ correspond to strengthening the model's assumptions, each $\delta \in FF$ satisfies $\Theta(\delta) \neq \emptyset$, whereas any $\delta' < \delta$ yields $\Theta(\delta') = \emptyset$.

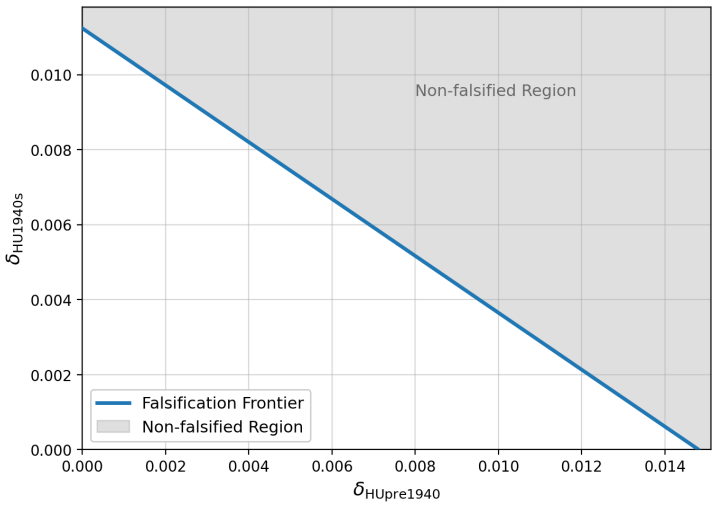
[2] Masten and Poirier (2021) note that this approach is particularly useful when one instrument is more credible than the other: when an instrument is likely to violate the exclusion restriction—that is, when it has a direct effect on the dependent variable—including it as a control, rather than omitting it, mitigates the resulting omitted-variable bias.

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3 Wide Replication Results

3.1 Falsification Frontier and Falsification Adaptive Set

Figure 1: Estimated falsification frontier



Notes: The x-axis indexes δ values for HUpre1940 variable, and the y-axis indexes δ values for HU1940s variable. The blue line is the estimated falsification frontier, and the shaded area is the non-falsified region.

We begin by estimating the FF in (4). Figure 1 plots the estimated FF for the main specification in Table 4 of Roche (2020). The blue line is the estimated FF, which traces the minimally non-falsified combinations of exclusion-restriction relaxations, and the shaded area is the estimated non-falsified region. Because $\delta_{\text{HUpre1940}}$ and δ_{HU1940s} are restricted to be nonnegative, the figure is confined to the first quadrant, and the baseline IV model corresponds to the origin.[3] The estimated frontier places the origin in the falsified region, so the identifying assumption underlying the original analysis—exclusion of both instruments—yields an empty identified set for θ at the point estimates. This means that exact exclusion of both instruments cannot hold jointly; to obtain any identified value of θ, the exclusion restriction on at least one instrument must be relaxed to a strictly positive δ_l. Specifically, imposing the exclusion restriction on HUpre1940 ($\delta_{\text{HUpre1940}} = 0$) requires relaxing it on HU1940s in order to avoid falsification, and

[3]The estimated FF is linear. In fact, when both IVs are relevant (i.e., $\pi_l \neq 0$ for $l = 1, 2$), the FF is always linear. See Appendix B.

vice versa. It is worth noting, however, that the *FF* only indicates whether falsification is robust to relaxations of the baseline model; it does not by itself quantify the degree to which any individual exclusion restriction is violated.[4] The *FF* characterizes the combinations of relaxations under which falsification is avoided, not the size of the violation attributable to either instrument.

While a conventional overidentification test does not reject the joint validity of the two instruments at standard levels in both Roche (2020) and our narrow replication (cf. Table A.4 in Appendix A), the estimated frontier nonetheless places the baseline model in the falsified region. These results can be reconciled by noting that the overidentification test may be inconsistent against certain alternatives (cf. Kadane and Anderson (1977); Small (2007); Guggenberger (2012)). This is also precisely where the approach of Masten and Poirier (2021) is informative.

Table 1: Estimated falsification adaptive set

	(1)	(2)
Excluded instruments	HUpre1940	HU1940s
Controlled just-identified IV estimate	0.0504***	-0.0096
	(0.0193)	(0.0450)
First-stage F-statistic	55.32	14.78
Falsification adaptive set	$[-0.0096, 0.0504]$	
Full controls	Yes	Yes
Commuting zone fixed effects	Yes	Yes
Controlled instruments	HU1940s	HUpre1940
Observations	95,097	95,097
Number of commuting zones	252	252

Notes: This table reports the second-stage 2SLS results for the relationship between road connectivity and patenting, using one instrument as the excluded instrument and including the other as a control. The outcome variable is the log number of granted U.S. patents applied for between 2011 and 2013 by assignees located in the block group, as in Roche (2020). The percentage of housing units in a given block group that were built before 1940 is denoted by HUpre1940, and the percentage of housing units built between 1940 and 1949 is denoted by HU1940s. Robust standard errors, clustered at the commuting-zone level, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We move on to estimation of the *FAS*. Table 1 displays the results. Column 1 uses HUpre1940 as the instrument while controlling for HU1940s, whereas Column 2 uses HU1940s as the in-

[4] If instrument-specific information is of interest, one can instead apply tests of instrument validity, such as Kitagawa (2015) and Kaido and Ponomarev (2025). Note, however, that they consider testing whether the exclusion restriction holds rather than quantifying the degree of any violation.

strument while controlling for HUpre1940. Both results are based on the specification reported in Column 6 of Table 4 in Roche (2020), including the same control variables and fixed effects.

The estimated FAS is $[-0.0096, 0.0504]$, suggesting that when we relax the exclusion restrictions on the instruments in the minimal way so that the model is non-falsified, the causal parameter θ lies between the two endpoints of the FAS. Since the FAS contains negative values, the original point estimate $\hat{\theta} = 0.0426 > 0$ in Roche (2020) is not robust to *all of the minimal relaxations* of the exclusion restrictions represented by the FF. That is, the sign of the effect is no longer determined under minimally non-falsified relaxations, so the qualitative result of Roche (2020) is not guaranteed. The FAS, however, pools the identified values across all minimally non-falsified relaxations. It contains negative values because the relaxations that maintain HU1940s as a valid excluded instrument yield negative estimates, and positive values because those that maintain HUpre1940 as a valid excluded instrument yield positive estimates. Its sign-indeterminacy therefore reflects disagreement between the two maintained assumptions rather than ambiguity within either. Falsification of the baseline model thus does not imply that the original conclusion is overturned; instead, the recovery of the positive effect depends on which exclusion restriction is maintained. Some minimally non-falsified relaxations preserve the positive conclusion of Roche (2020), whereas others do not.

The reason for this result is an asymmetry across the two instruments, which reflects differences in both the empirical informativeness of the instruments and the plausibility of the maintained exclusion restrictions. We first consider instrument relevance. Both controlled just-identified specifications exceed the weak-instrument thresholds in Staiger and Stock (1997) and Stock and Yogo (2005), indicating that weak identification is unlikely to drive the results. Nevertheless, the specification that uses HU1940s as the excluded instrument exhibits substantially weaker first-stage relevance ($F = 14.78$), resulting in a considerably less precise estimate. In contrast, when HU1940s is treated as a control variable and HUpre1940 is used as the instrument, the first-stage F-statistic is large ($F = 55.32$), and the estimated coefficient remains positive and is similar in magnitude to the original estimate reported in Roche (2020).

This asymmetry in relevance is consistent with the morphological evidence in Boeing (2021), who treats 1940 as a "rupture point" in American street-network development. Boeing (2021) documents that tracts developed after 1939 are systematically less grid-like than those developed before 1940. Since the grid-like street structure is the source of the road connectivity that Roche (2020) instruments for, HUpre1940 captures this connectivity mechanism

more directly than HU1940s. This provides a substantive explanation for the first-stage pattern documented above, in which HUpre1940 is the considerably stronger instrument.[5]

We also reconsider the institutional validity of the instruments. Roche (2020) notes that many U.S. cities experienced substantial demographic and urban restructuring around and after World War II. We infer from this that the housing stock constructed during the 1940s may be more directly correlated with wartime industrial development and subsequent urban expansion, which would provide a plausible reason why HU1940s could affect contemporary innovation through channels other than road connectivity. The persistence of spatial composition keeps both historical shares correlated with contemporary density, and hence relevant as instruments; the reason HU1940s is different is its association with wartime industrial development, which makes its exclusion restriction relatively less plausible. This argument does not establish invalidity, but it makes the maintained exclusion restriction for HU1940s relatively less compelling than that for HUpre1940, which is less correlated with the wartime era.

Moreover, Boeing (2021) documents that, beginning around 1940, massive state intervention rebuilt mobility infrastructure around the technological and spatial logic of the automobile, reshaping the form of American urbanization. This period of state-driven infrastructural and industrial transformation likely affected the long-run economic trajectory of neighborhoods through channels beyond road connectivity—for instance, through the location of wartime and postwar industrial activity, which is plausibly correlated with 1940s residential development and may bear directly on subsequent innovation. As a result, HU1940s may affect the outcome variable through a direct channel, which would violate the exclusion restriction for this variable.

Therefore, although both controlled just-identified estimators formally contribute to the *FAS*, the recovery of the qualitative conclusion of Roche (2020) is asymmetric across the two maintained assumptions. Conditional on maintaining HUpre1940 as the valid excluded instrument, the estimated effect remains positive and similar in magnitude to the original estimate in Roche (2020). By contrast, maintaining HU1940s as the excluded instrument no longer preserves the qualitative conclusion. Since the *FF* itself does not identify which exclusion restriction is violated, the data alone cannot distinguish between these two maintained assumptions. Nevertheless, the historical and institutional discussion above provides a plausible reason for

[5]We emphasize, however, that first-stage strength is related to relevance, not to the exclusion restriction: the weaker relevance of HU1940s bears on the precision of that specification, not on its validity. The case for treating HUpre1940 as the maintained exclusion rests on the institutional argument below, not on its stronger first stage.

placing greater weight on the former interpretation. Therefore, conditional on the credibility of HUpre1940 as the maintained excluded instrument—for which we have given historical and institutional reasons—the positive effect of connectivity on innovation is preserved, though we emphasize that this conclusion rests on an exclusion restriction that the data cannot verify.

3.2 Breakdown Analysis along the Falsification Frontier

As shown before, the estimation result depends on which instrument violates the exclusion restriction, yielding an asymmetry in the recovery of the original result in Roche (2020) across instruments. To further investigate this asymmetry, we consider a breakdown frontier analysis. Breakdown frontier analysis answers the question of what the weakest assumptions are needed to draw a prespecified conclusion about a parameter (Masten and Poirier (2020)). We seek the set of δ values such that $\inf \Theta(\delta) \geq 0$ *and* the model is not falsified. To this end, we define the "conclusion-preserving region" as follows.

Definition 3.1 (Conclusion-Preserving Region). *For a given δ, let $\Theta(\delta)$ denote the identified set for θ. The conclusion-preserving region for* $\inf \Theta(\delta) \geq 0$, *denoted by FF^+, is*

$$FF^+ \equiv \{\delta \in FF \,:\, \inf \Theta(\delta) \geq 0\}. \tag{6}$$

It is worth noting that the conclusion-preserving region differs from the breakdown frontier of Masten and Poirier (2020). While the breakdown frontier is defined over the full space of relaxations, FF^+ is the conclusion-preserving subset *of the falsification frontier*: it collects the values of δ on the falsification frontier such that the conclusion $\inf \Theta(\delta) \geq 0$ holds.

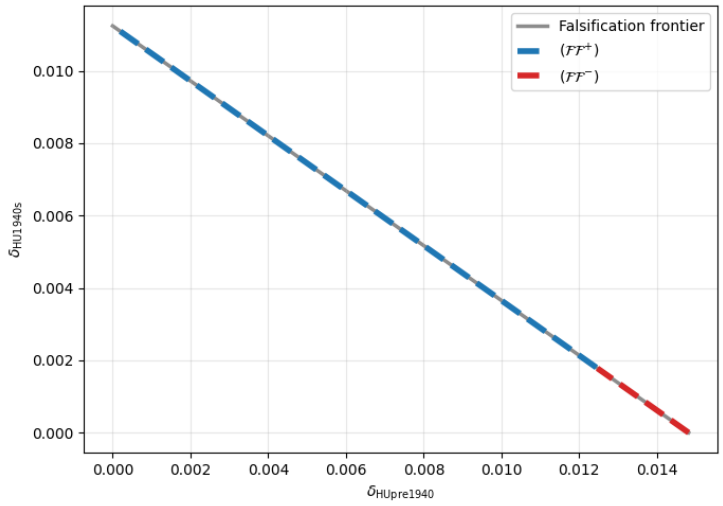
In Figure 2, the blue line corresponds to FF^+ and the red line corresponds to $FF^- \equiv FF \setminus FF^+$. The boundary between the two regions is exactly where $\theta = 0$, which corresponds to $\hat{\delta}_0 \equiv \left(|\hat{\psi}_1|, |\hat{\psi}_2|\right) = (|0.0124|, |-0.0017|) = (0.0124, 0.0017)$.[6]

Figure 2 provides several interesting results. First, maintaining the exclusion restriction for HUpre1940 while relaxing the exclusion restriction on HU1940s preserves the main empirical conclusion of Roche (2020) while the model is not falsified. This is the same as what we have seen in Table 1, and the blue line in Figure 2 contains the point whose first coordinate is zero (i.e., the exclusion restriction for HUpre1940 is imposed).

Second, the main conclusion holds even when we relax both exclusion restrictions simulta-

[6] We show how to find this value in Appendix B.

Figure 2: Conclusion-Preserving Region



Notes: The x-axis indexes δ values for HUpre1940 variable, and the y-axis indexes δ values for HU1940s variable. The gray solid line is the estimated falsification frontier. The blue dashed line is the set of δ such that $\theta \geq 0$, and the red dashed line depicts the set of δ such that $\theta < 0$.

neously. The blue line in Figure 2 contains not only the point whose first coordinate is zero, but also points whose first coordinate is nonzero. This implies that when we relax the exclusion restrictions on the two instruments in an appropriate sense, the model is not falsified and the estimated effect remains non-negative, so the main conclusion of Roche (2020) holds. Recovery of the positive conclusion therefore does not require holding either instrument exactly valid. Moreover, this conclusion-preserving region covers a large portion of the FF. The blue line is evidently longer than the red line, so the estimated effect remains non-negative—and the model non-falsified—over most of the minimal relaxations of the exclusion restrictions on both instruments.[7] The difference in length between the blue and red lines also reflects the asymmetric effect of violations of the exclusion restriction on the recovery of the main conclusion.

4 Conclusion

We replicate the empirical results of Roche (2020) and evaluate the identifying content of the original IV design under relaxed exclusion restrictions, using the framework of Masten and Poirier (2021). We find that the baseline two-instrument model is falsified, and we estimate the

[7] The length of the blue line is 0.016, whereas the length of the red line is 0.003.

FF and *FAS*. The original sign conclusion in Roche (2020) is not preserved over all minimally non-falsified relaxations of the exclusion restrictions; however, it holds over a large portion of them. In particular, using our preferred IV as the excluded instrument and the other as a control yields a positive estimated effect of road connectivity on innovation. We also explain why one IV is preferred over the other, drawing on the urban-planning literature. More broadly, this paper illustrates how to conduct sensitivity analysis of the exclusion restriction when multiple instruments are available for a single endogenous treatment, including how to assess which instrument to maintain.

Disclosure Statement

The authors used ChatGPT for language editing and for limited assistance with Python coding. The authors reviewed and verified all AI-generated output and take full responsibility for the content of the published article.

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A Narrow Replication Results

In this section, we report the narrow replication results. We follow the original data and do-file specifications. All computations in this paper, including both the narrow and wide replications and the construction of all tables and figures, are conducted in Python, whereas Roche (2020) conducts the original analysis in Stata. Specifically, we used Python's PyFixest package to compute fixed effect regression. Tables A.1–A.6 provide narrow replications results reported in *Tables* 1–6 of Roche (2020). Except for small numerical discrepancies in the first stage F-statistics, our narrow replication reproduces the original results. These discrepancies arise from differences between Stata and Python (PyFixest) in the finite sample, and do not affect the point estimates or their statistical significance.

Table A.1: Replication Results of Table 1 in Roche (2020)

Census Block Group Level	Mean	Min	p25	p50	p95	Max
Innovation						
Number of Patents (2011-2013)	0.18	0.00	0.00	0.00	0.00	1025.00
Patent (= 0/1)	0.04	0.00	0.00	0.00	0.00	1.00
Knowledge Exchange						
Total Same BG Citations	0.03	0.00	0.00	0.00	0.00	612.00
Self Same BG Citations	0.03	0.00	0.00	0.00	0.00	607.00
Non-Self Same BG Citations	0.002	0.00	0.00	0.00	0.00	33.00
Physical Network Structure						
Connectivity (in miles per square mile)	2.83	0.00	0.86	2.20	7.94	466.98
HUpre1940	0.20	0.00	0.00	0.06	0.73	1.00
HU1940s	0.08	0.00	0.00	0.04	0.31	1.00
Social Activity						
Number of Bars, Restaurants, and Hotels	2.54	0.00	1.00	2.00	6.00	143.00
Formal Knowledge						
Campus	0.007	0.00	0.00	0.00	0.00	1.00
Human Capital						
Accessibility (in thousands)	292.91	0.00	97.74	194.05	1039.11	1598.20
Employment 2010 (in thousands)	0.55	0.00	0.05	0.16	2.03	232.46
Employment 2005 (in thousands)	0.42	0.00	0.00	0.07	1.70	167.37
Inventors 2005	0.02	0.00	0.00	0.00	0.00	18.00
Inventors 2000	0.02	0.00	0.00	0.00	0.00	10.00
Sociodemographic (in thousands)						
Population 2010	1.32	0.00	0.88	1.19	2.48	19.51
Population 2000	1.28	0.00	0.88	1.17	2.33	12.78
Physical Geography (in hundred acres)						
Area Water	0.05	0.00	0.00	0.00	0.23	5.47
Area Developable Land	1.79	0.00	0.73	1.38	4.85	6.40
Area Land	1.86	0.000	0.77	1.45	5.02	6.40
Observations				122,899		

Table A.2: Replication Results of Table 2 in Roche (2020)

DV: Number of Patents (log)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Connectivity (log)	0.0164***	0.0149***	0.0158***	0.00426**	0.0174***	0.0154***	0.0151***	0.0203***	0.00452***	0.00425***
	(0.00289)	(0.00243)	(0.00281)	(0.00184)	(0.00305)	(0.00260)	(0.00257)	(0.00342)	(0.00129)	(0.00140)
No. Bars		0.0120***							0.00361***	0.00381***
		(0.00132)							(0.00107)	(0.00103)
Campus			0.2238***						0.0982***	0.0996***
			(0.0394)						(0.0227)	(0.0227)
Employment 2005				0.00214					0.00417	0.00400
				(0.00707)					(0.00617)	(0.00517)
Employment 2010				0.0393***					0.0369***	0.0368***
				(0.00696)					(0.00551)	(0.00492)
Population 2000					-0.0277***				-0.0227***	-0.0235***
					(0.00688)				(0.00717)	(0.00697)
Population 2010					0.0389***				0.0133**	0.0121*
					(0.00695)				(0.00602)	(0.00618)
Accessibility						0.000034*			0.000035**	0.000025
						(0.000020)			(0.000016)	(0.000022)
Inventors 2000							-0.00224		-0.00498	-0.00437
							(0.00324)		(0.00327)	(0.00377)
Inventors 2005							-0.000939		-0.000153	0.00107
							(0.00344)		(0.00333)	(0.00344)
College Degree 2000							-0.0282***		-0.00384	-0.00716
							(0.00878)		(0.00657)	(0.00745)
College Degree 2010							0.0500***		0.0200***	0.0182***
							(0.00903)		(0.00662)	(0.00656)
Area Water								0.0191***	0.0139***	0.0163***
								(0.00386)	(0.00488)	(0.00414)
Area Developable Land								0.0132***	0.00632	0.00882**
								(0.00315)	(0.0038)	(0.00390)
Area Land								0.00415	0.00360	0.00282
								(0.00299)	(0.00316)	(0.00335)
Observations	121,398	119,159	121,398	121,398	121,398	121,398	96,973	121,398	95,294	95,207
R-squared	0.00183	0.0198	0.00715	0.0938	0.00485	0.00224	0.00840	0.00996	0.0996	0.0970
Fixed Effects	czone	czone	czone	czone	czone	czone	czone	czone	czone	county
Number of Groups	261	257	261	261	261	261	257	261	253	972

Notes: Robust standard errors, clustered at the commuting-zone level in columns 1–9 and at the county level in column 10, are in parentheses. Employment and population measures are in thousands. Geographic area variables are in hundreds. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Replication Results of Table 3 in Roche (2020)

DV: Patent (= 0/1)	(1)	(2)	(3)
Connectivity (log)	0.0108***	0.00492***	0.00466***
	(0.00179)	(0.00115)	(0.00104)
Social Activity Controls	No	Yes	Yes
Formal Knowledge Controls	No	Yes	Yes
Human Capital Controls	No	Yes	Yes
Sociodemographic Controls	No	Yes	Yes
Physical Geography Controls	No	Yes	Yes
Observations	121,398	95,294	95,207
R-squared	0.00148	0.0760	0.0712
Fixed Effects	czone	czone	county
Number of Groups	261	253	972
Std. Errors	Robust	Robust	Robust
Log Likelihood	30,061.2	27,711.2	28,274.2

Notes: Robust standard errors, clustered at the commuting-zone level in columns 1–2 and at the county level in column 3, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Replication Results of Table 4 in Roche (2020)

DV: Number of Patents (log)	(1)	(2)	(3)	(4)	(5)	(6)
Second Stage						
Connectivity (log)	0.0965***	0.0915***	0.0799***	0.0517***	0.0504***	0.0426**
	(0.0184)	(0.0176)	(0.0163)	(0.0196)	(0.0192)	(0.0188)
Social Activity Controls	No	Yes	Yes	Yes	Yes	Yes
Formal Knowledge Controls	No	Yes	Yes	Yes	Yes	Yes
Human Capital Controls	No	No	No	Yes	Yes	Yes
Sociodemographic Controls	No	No	Yes	No	Yes	Yes
Physical Geography Controls	Yes	Yes	Yes	Yes	Yes	Yes
First Stage						
HUpre1940	0.2928***	0.2926***	0.2874***	0.2475***	0.2468***	0.2467***
	(0.0359)	(0.0368)	(0.0362)	(0.0345)	(0.0342)	(0.0332)
HU1940s						0.1872***
						(0.0487)
Observations	120,926	118,838	118,838	95,097	95,097	95,097
First Stage F-statistics	66.71	63.22	62.92	51.43	52.09	32.80
Hansen's J Stat. p-value						0.212
Fixed Effects	czone	czone	czone	czone	czone	czone
Number of Groups	260	256	256	252	252	252

Notes: Columns 1–6 report the replication results. In columns 1–5, HUpre1940 is used as a sole IV. In column 6, HUpre1940 and HU1940s are both used as IVs. Robust standard errors are in parentheses and clustered at the commuting-zone level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Replication Results of Table 5 in Roche (2020)

	Non-Self				Self			
DV: Number Citations (log)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Connectivity (log)	0.000540***	0.000213**	0.00431**	0.00377**	0.00316***	0.000759*	-0.000241	-0.00549
	(0.000152)	(0.000103)	(0.00211)	(0.00187)	(0.000796)	(0.000417)	(0.00963)	(0.00748)
Number of Patents (log)				0.0220***				0.2138***
				(0.00485)				(0.0203)
Social Activity Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Formal Knowledge Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Human Capital Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Sociodemographic Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Physical Geography Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model	OLS	OLS	IV	IV	OLS	OLS	IV	IV
First Stage								
HUpre1940			0.1853***	0.1851***			0.1853***	0.1851***
			(0.0332)	(0.0332)			(0.0332)	(0.0332)
HU1940s			0.2000***	0.2002***			0.2000***	0.2002***
			(0.0435)	(0.0435)			(0.0435)	(0.0435)
Number of Patents (log)			No	Yes			No	Yes
Other Controls			Yes	Yes			Yes	Yes
First Stage F-statistics			25.48	25.57			25.48	25.57
Hansen's J Stat. p-value			0.905	0.636			0.667	0.744
Observations	121,398	119,142	95,097	95,097	121,398	119,142	95,097	95,097
R-Squared	0.00047	0.00322	-0.00651	0.0340	0.00141	0.00815	0.00811	0.2967
Fixed Effects	czone	czone	czone	czone	czone	czone	czone	czone
Number of Groups	261	257	252	252	261	257	252	252
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust

Notes: Columns 1, 2, 5, and 6 report the OLS estimation results. Columns 3, 4, 7, and 8 present the IV estimation results using HUpre1940 and HU1940s. Robust standard errors are in parentheses and clustered at the commuting-zone level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Replication Results of Table 6 in Roche (2020)

DV: Number of Patents (log)	(1)	(2)	(3)	(4)	(5)	(6)
Connectivity (log)	0.0163***	0.00357**	0.00336***	0.00187*	0.0124***	0.00239*
	(0.00292)	(0.00146)	(0.000940)	(0.00106)	(0.00217)	(0.00123)
High Population = 1	-0.0202***	-0.00525				
	(0.00548)	(0.00420)				
High Population = 1 × Connectivity (log)	0.0209***	0.00881**				
	(0.00522)	(0.00411)				
High Employment = 1			0.0911***	0.0611***		
			(0.0172)	(0.0172)		
High Employment = 1 × Connectivity (log)			0.0590***	0.0274**		
			(0.0139)	(0.0129)		
High No. Bars = 1					-0.00619	-0.00299
					(0.0174)	(0.00778)
High No. Bars = 1 × Connectivity (log)					0.0813***	0.0360***
					(0.0150)	(0.00835)
Other Social Activity Controls	No	Yes	No	Yes	No	No
Formal Knowledge Controls	No	Yes	No	Yes	No	Yes
Other Human Capital Controls	No	Yes	No	Yes	No	Yes
Other Sociodemographic Controls	No	Yes	No	Yes	No	Yes
Physical Geography	Yes	Yes	Yes	Yes	Yes	Yes
Observations	121,398	95,294	121,398	95,294	119,159	95,294
R-squared	0.0105	0.0991	0.0551	0.0855	0.0217	0.0999
Fixed Effects	czone	czone	czone	czone	czone	czone
Number of Groups	261	253	261	253	257	253

Notes: The variable 'High Population' is equal to one if population in 2010 is above 1,650. The variable 'High Employment' is equal to one if employment in 2010 is above 950. The variable 'High No. Bars' is equal to one if bars, restaurants, and hotels are above 5. Robust standard errors are in parentheses and clustered at the commuting-zone level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Auxiliary Mathematical Results

We first show that the FF is linear in our case. Without loss of generality, assume that $\min_{\ell=1,2\ :\pi_\ell\neq 0} \frac{\psi_\ell}{\pi_\ell} = \frac{\psi_2}{\pi_2}$ and $\max_{\ell=1,2\ :\pi_\ell\neq 0} \frac{\psi_\ell}{\pi_\ell} = \frac{\psi_1}{\pi_1}$. Then, $\delta_1 = |\psi_1 - \theta\pi_1| = \psi_1 - \theta\pi_1$ and $\delta_2 = -\psi_2 + \theta\pi_2$ for all $\theta \in \left[\frac{\psi_2}{\pi_2}, \frac{\psi_1}{\pi_1}\right]$. Therefore, we obtain

$$\delta_2 = -\frac{\pi_2}{\pi_1}\delta_1 + \frac{\psi_1\pi_2}{\pi_1} - \psi_2, \tag{A.1}$$

provided $\pi_1 \neq 0$ and $\pi_2 \neq 0$. That is, if two instruments are relevant, then FF is linear on $\mathbb{R}^2$.

We then characterize the form of the conclusion-preserving region, FF^+, defined in (6). From the definition of the FF and using Lemma 2 in Masten and Poirier (2021) that $\Theta(\delta(\theta)) = \theta$ for all $\theta \in \left[\min_{\ell=1,2\ :\pi_\ell\neq 0} \frac{\psi_\ell}{\pi_\ell}, \max_{\ell=1,2\ :\pi_\ell\neq 0} \frac{\psi_\ell}{\pi_\ell}\right]$, where $\delta(\theta) = (|\psi_1 - \theta\pi_1|, |\psi_2 - \theta\pi_2|)$, we can show that FF^+ collapses to the following form:

$$FF^+ = \left\{ \delta \in FF \ : \ \delta_\ell = |\psi_\ell - b\pi_\ell|, \ \ \ell = 1,2, \ \ b \in \left[0, \max_{\ell=1,2\ :\pi_\ell\neq 0} \frac{\psi_\ell}{\pi_\ell}\right] \right\}. \tag{A.2}$$

Therefore, the value δ_0 corresponding to $b = 0$ is $\delta_0 = (|\psi_1|, |\psi_2|)$. To estimate δ_0, we replace $|\psi_1|$ and $|\psi_2|$ with their consistent estimators.